

4 Statistics

4.1 Introduction

In Chapter 2–Chapter 3, we assumed all the parameters θ of our probability models were known. In this chapter, we discuss how to learn these parameters from data.

The process of estimating θ from $\mathcal{D}$ is called **model fitting**, or **training**, and is at the heart of machine learning. There are many methods for producing such estimates, but most boil down to an optimization problem of the form

$$\hat{\theta} = \underset{\theta}{\operatorname{argmin}} \mathcal{L}(\theta) \quad (4.1)$$

where $\mathcal{L}(\theta)$ is some kind of loss function or objective function. We discuss several different loss functions in this chapter. In some cases, we also discuss how to solve the optimization problem in closed form. In general, however, we will need to use some kind of generic optimization algorithm, which we discuss in Chapter 8.

In addition to computing a **point estimate**, $\hat{\theta}$, we discuss how to model our uncertainty or confidence in this estimate. In statistics, the process of quantifying uncertainty about an unknown quantity estimated from a finite sample of data is called **inference**. We will discuss both Bayesian and frequentist approaches to inference.¹

4.2 Maximum likelihood estimation (MLE)

The most common approach to parameter estimation is to pick the parameters that assign the highest probability to the training data; this is called **maximum likelihood estimation** or **MLE**. We give more details below, and then give a series of worked examples.

4.2.1 Definition

We define the MLE as follows:

$$\hat{\theta}_{\text{mle}} \triangleq \underset{\theta}{\operatorname{argmax}} p(\mathcal{D}|\theta) \quad (4.2)$$

1. In the deep learning community, the term “inference” refers to what we will call “prediction”, namely computing $p(y|\mathbf{x}, \hat{\theta})$.

We usually assume the training examples are independently sampled from the same distribution, so the (conditional) likelihood becomes

$$p(\mathcal{D}|\boldsymbol{\theta}) = \prod_{n=1}^N p(\mathbf{y}_n|\mathbf{x}_n, \boldsymbol{\theta}) \quad (4.3)$$

This is known as the **iid** assumption, which stands for “independent and identically distributed”. We usually work with the **log likelihood**, which is given by

$$\ell(\boldsymbol{\theta}) \triangleq \log p(\mathcal{D}|\boldsymbol{\theta}) = \sum_{n=1}^N \log p(\mathbf{y}_n|\mathbf{x}_n, \boldsymbol{\theta}) \quad (4.4)$$

This decomposes into a sum of terms, one per example. Thus the MLE is given by

$$\hat{\boldsymbol{\theta}}_{\text{mle}} = \underset{\boldsymbol{\theta}}{\operatorname{argmax}} \sum_{n=1}^N \log p(\mathbf{y}_n|\mathbf{x}_n, \boldsymbol{\theta}) \quad (4.5)$$

Since most optimization algorithms (such as those discussed in Chapter 8) are designed to *minimize* cost functions, we can redefine the **objective function** to be the (conditional) **negative log likelihood** or **NLL**:

$$\text{NLL}(\boldsymbol{\theta}) \triangleq -\log p(\mathcal{D}|\boldsymbol{\theta}) = -\sum_{n=1}^N \log p(\mathbf{y}_n|\mathbf{x}_n, \boldsymbol{\theta}) \quad (4.6)$$

Minimizing this will give the MLE. If the model is unconditional (unsupervised), the MLE becomes

$$\hat{\boldsymbol{\theta}}_{\text{mle}} = \underset{\boldsymbol{\theta}}{\operatorname{argmin}} -\sum_{n=1}^N \log p(\mathbf{y}_n|\boldsymbol{\theta}) \quad (4.7)$$

since we have outputs $\mathbf{y}_n$ but no inputs $\mathbf{x}_n$.²

Alternatively we may want to maximize the *joint* likelihood of inputs and outputs. The MLE in this case becomes

$$\hat{\boldsymbol{\theta}}_{\text{mle}} = \underset{\boldsymbol{\theta}}{\operatorname{argmin}} -\sum_{n=1}^N \log p(\mathbf{y}_n, \mathbf{x}_n|\boldsymbol{\theta}) \quad (4.8)$$

4.2.2 Justification for MLE

There are several ways to justify the method of MLE. One way is to view it as simple point approximation to the Bayesian posterior $p(\boldsymbol{\theta}|\mathcal{D})$ using a uniform prior, as explained in Section 4.6.7.1.

2. In statistics, it is standard to use $\mathbf{y}$ to represent variables whose generative distribution we choose to model, and to use $\mathbf{x}$ to represent exogenous inputs which are given but not generated. Thus supervised learning concerns fitting conditional models of the form $p(\mathbf{y}|\mathbf{x})$, and unsupervised learning is the special case where $\mathbf{x} = \emptyset$, so we are just fitting the unconditional distribution $p(\mathbf{y})$. In the ML literature, supervised learning treats $\mathbf{y}$ as generated and $\mathbf{x}$ as given, but in the unsupervised case, it often switches to using $\mathbf{x}$ to represent generated variables.

In particular, suppose we approximate the posterior by a delta function, $p(\boldsymbol{\theta}|\mathcal{D}) = \delta(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}_{\text{map}})$, where $\hat{\boldsymbol{\theta}}_{\text{map}}$ is the posterior mode, given by

$$\hat{\boldsymbol{\theta}}_{\text{map}} = \underset{\boldsymbol{\theta}}{\operatorname{argmax}} \log p(\boldsymbol{\theta}|\mathcal{D}) = \underset{\boldsymbol{\theta}}{\operatorname{argmax}} \log p(\mathcal{D}|\boldsymbol{\theta}) + \log p(\boldsymbol{\theta}) \quad (4.9)$$

If we use a uniform prior, $p(\boldsymbol{\theta}) \propto 1$, the MAP estimate becomes equal to the MLE, $\hat{\boldsymbol{\theta}}_{\text{map}} = \hat{\boldsymbol{\theta}}_{\text{mle}}$.

Another way to justify the use of the MLE is that the resulting predictive distribution $p(\mathbf{y}|\hat{\boldsymbol{\theta}}_{\text{mle}})$ is as close as possible (in a sense to be defined below) to the **empirical distribution** of the data. In the unconditional case, the empirical distribution is defined by

$$p_{\mathcal{D}}(\mathbf{y}) \triangleq \frac{1}{N} \sum_{n=1}^N \delta(\mathbf{y} - \mathbf{y}_n) \quad (4.10)$$

We see that the empirical distribution is a series of delta functions or “spikes” at the observed training points. We want to create a model whose distribution $q(\mathbf{y}) = p(\mathbf{y}|\boldsymbol{\theta})$ is similar to $p_{\mathcal{D}}(\mathbf{y})$.

A standard way to measure the (dis)similarity between probability distributions p and q is the **Kullback Leibler divergence**, or **KL divergence**. We give the details in Section 6.2, but in brief this is defined as

$$D_{\text{KL}}(p \parallel q) = \sum_{\mathbf{y}} p(\mathbf{y}) \log \frac{p(\mathbf{y})}{q(\mathbf{y})} \quad (4.11)$$

$$= \underbrace{\sum_{\mathbf{y}} p(\mathbf{y}) \log p(\mathbf{y})}_{-\mathbb{H}(p)} - \underbrace{\sum_{\mathbf{y}} p(\mathbf{y}) \log q(\mathbf{y})}_{\mathbb{H}_{ce}(p, q)} \quad (4.12)$$

where $\mathbb{H}(p)$ is the entropy of p (see Section 6.1), and $\mathbb{H}_{ce}(p, q)$ is the cross-entropy of p and q (see Section 6.1.2). One can show that $D_{\text{KL}}(p \parallel q) \geq 0$, with equality iff $p = q$.

If we define $q(\mathbf{y}) = p(\mathbf{y}|\boldsymbol{\theta})$, and set $p(\mathbf{y}) = p_{\mathcal{D}}(\mathbf{y})$, then the KL divergence becomes

$$D_{\text{KL}}(p \parallel q) = \sum_{\mathbf{y}} [p_{\mathcal{D}}(\mathbf{y}) \log p_{\mathcal{D}}(\mathbf{y}) - p_{\mathcal{D}}(\mathbf{y}) \log q(\mathbf{y})] \quad (4.13)$$

$$= -\mathbb{H}(p_{\mathcal{D}}) - \frac{1}{N} \sum_{n=1}^N \log p(\mathbf{y}_n|\boldsymbol{\theta}) \quad (4.14)$$

$$= \text{const} + \text{NLL}(\boldsymbol{\theta}) \quad (4.15)$$

The first term is a constant which we can ignore, leaving just the NLL. Thus minimizing the KL is equivalent to minimizing the NLL which is equivalent to computing the MLE, as in Equation (4.7).

We can generalize the above results to the supervised (conditional) setting by using the following empirical distribution:

$$p_{\mathcal{D}}(\mathbf{x}, \mathbf{y}) = p_{\mathcal{D}}(\mathbf{y}|\mathbf{x})p_{\mathcal{D}}(\mathbf{x}) = \frac{1}{N} \sum_{n=1}^N \delta(\mathbf{x} - \mathbf{x}_n) \delta(\mathbf{y} - \mathbf{y}_n) \quad (4.16)$$

The expected KL then becomes

$$\mathbb{E}_{p_{\mathcal{D}}(\mathbf{x})} [D_{\text{KL}}(p_{\mathcal{D}}(Y|\mathbf{x}) \parallel q(Y|\mathbf{x}))] = \sum_{\mathbf{x}} p_{\mathcal{D}}(\mathbf{x}) \left[\sum_{\mathbf{y}} p_{\mathcal{D}}(\mathbf{y}|\mathbf{x}) \log \frac{p_{\mathcal{D}}(\mathbf{y}|\mathbf{x})}{q(\mathbf{y}|\mathbf{x})} \right] \quad (4.17)$$

$$= \text{const} - \sum_{\mathbf{x}, \mathbf{y}} p_{\mathcal{D}}(\mathbf{x}, \mathbf{y}) \log q(\mathbf{y}|\mathbf{x}) \quad (4.18)$$

$$= \text{const} - \frac{1}{N} \sum_{n=1}^N \log p(\mathbf{y}_n | \mathbf{x}_n, \boldsymbol{\theta}) \quad (4.19)$$

Minimizing this is equivalent to minimizing the conditional NLL in Equation (4.6).

4.2.3 Example: MLE for the Bernoulli distribution

Suppose Y is a random variable representing a coin toss, where the event $Y = 1$ corresponds to heads and $Y = 0$ corresponds to tails. Let $\theta = p(Y = 1)$ be the probability of heads. The probability distribution for this rv is the Bernoulli, which we introduced in Section 2.4.

The NLL for the Bernoulli distribution is given by

$$\text{NLL}(\theta) = -\log \prod_{n=1}^N p(y_n | \theta) \quad (4.20)$$

$$= -\log \prod_{n=1}^N \theta^{\mathbb{I}(y_n=1)} (1-\theta)^{\mathbb{I}(y_n=0)} \quad (4.21)$$

$$= -\sum_{n=1}^N [\mathbb{I}(y_n = 1) \log \theta + \mathbb{I}(y_n = 0) \log(1 - \theta)] \quad (4.22)$$

$$= -[N_1 \log \theta + N_0 \log(1 - \theta)] \quad (4.23)$$

where we have defined $N_1 = \sum_{n=1}^N \mathbb{I}(y_n = 1)$ and $N_0 = \sum_{n=1}^N \mathbb{I}(y_n = 0)$, representing the number of heads and tails. (The NLL for the binomial is the same as for the Bernoulli, modulo an irrelevant $\binom{N}{c}$ term, which is a constant independent of θ .) These two numbers are called the **sufficient statistics** of the data, since they summarize everything we need to know about $\mathcal{D}$. The total count, $N = N_0 + N_1$, is called the **sample size**.

The MLE can be found by solving $\frac{d}{d\theta} \text{NLL}(\theta) = 0$. The derivative of the NLL is

$$\frac{d}{d\theta} \text{NLL}(\theta) = \frac{-N_1}{\theta} + \frac{N_0}{1-\theta} \quad (4.24)$$

and hence the MLE is given by

$$\hat{\theta}_{\text{mle}} = \frac{N_1}{N_0 + N_1} \quad (4.25)$$

We see that this is just the empirical fraction of heads, which is an intuitive result.

4.2.4 Example: MLE for the categorical distribution

Suppose we roll a K -sided dice N times. Let $Y_n \in \{1, \dots, K\}$ be the n 'th outcome, where $Y_n \sim \text{Cat}(\boldsymbol{\theta})$. We want to estimate the probabilities $\boldsymbol{\theta}$ from the dataset $\mathcal{D} = \{y_n : n = 1 : N\}$. The NLL is given by

$$\text{NLL}(\boldsymbol{\theta}) = - \sum_k N_k \log \theta_k \quad (4.26)$$

where N_k is the number of times the event $Y = k$ is observed. (The NLL for the multinomial is the same, up to irrelevant scale factors.)

To compute the MLE, we have to minimize the NLL subject to the constraint that $\sum_{k=1}^K \theta_k = 1$. To do this, we will use the method of Lagrange multipliers (see Section 8.5.1).³

The Lagrangian is as follows:

$$\mathcal{L}(\boldsymbol{\theta}, \lambda) \triangleq - \sum_k N_k \log \theta_k - \lambda \left(1 - \sum_k \theta_k \right) \quad (4.27)$$

Taking derivatives with respect to λ yields the original constraint:

$$\frac{\partial \mathcal{L}}{\partial \lambda} = 1 - \sum_k \theta_k = 0 \quad (4.28)$$

Taking derivatives with respect to θ_k yields

$$\frac{\partial \mathcal{L}}{\partial \theta_k} = -\frac{N_k}{\theta_k} + \lambda = 0 \implies N_k = \lambda \theta_k \quad (4.29)$$

We can solve for λ using the sum-to-one constraint:

$$\sum_k N_k = N = \lambda \sum_k \theta_k = \lambda \quad (4.30)$$

Thus the MLE is given by

$$\hat{\theta}_k = \frac{N_k}{\lambda} = \frac{N_k}{N} \quad (4.31)$$

which is just the empirical fraction of times event k occurs.

4.2.5 Example: MLE for the univariate Gaussian

Suppose $Y \sim \mathcal{N}(\mu, \sigma^2)$ and let $\mathcal{D} = \{y_n : n = 1 : N\}$ be an iid sample of size N . We can estimate the parameters $\boldsymbol{\theta} = (\mu, \sigma^2)$ using MLE as follows. First, we derive the NLL, which is given by

$$\text{NLL}(\mu, \sigma^2) = - \sum_{n=1}^N \log \left[\left(\frac{1}{2\pi\sigma^2} \right)^{\frac{1}{2}} \exp \left(-\frac{1}{2\sigma^2} (y_n - \mu)^2 \right) \right] \quad (4.32)$$

$$= \frac{1}{2\sigma^2} \sum_{n=1}^N (y_n - \mu)^2 + \frac{N}{2} \log(2\pi\sigma^2) \quad (4.33)$$

3. We do not need to explicitly enforce the constraint that $\theta_k \geq 0$ since the gradient of the Lagrangian has the form $-N_k/\theta_k - \lambda$; so negative values of θ_k would increase the objective, rather than minimize it. (Of course, this does not preclude setting $\theta_k = 0$, and indeed this is the optimal solution if $N_k = 0$.)

The minimum of this function must satisfy the following conditions, which we explain in Section 8.1.1.1:

$$\frac{\partial}{\partial \mu} \text{NLL}(\mu, \sigma^2) = 0, \quad \frac{\partial}{\partial \sigma^2} \text{NLL}(\mu, \sigma^2) = 0 \quad (4.34)$$

So all we have to do is to find this stationary point. Some simple calculus (Exercise 4.1) shows that the solution is given by the following:

$$\hat{\mu}_{\text{mle}} = \frac{1}{N} \sum_{n=1}^N y_n = \bar{y} \quad (4.35)$$

$$\hat{\sigma}_{\text{mle}}^2 = \frac{1}{N} \sum_{n=1}^N (y_n - \hat{\mu}_{\text{mle}})^2 = \frac{1}{N} \sum_{n=1}^N [y_n^2 + \hat{\mu}_{\text{mle}}^2 - 2y_n \hat{\mu}_{\text{mle}}] = s^2 - \bar{y}^2 \quad (4.36)$$

$$s^2 \triangleq \frac{1}{N} \sum_{n=1}^N y_n^2 \quad (4.37)$$

The quantities $\bar{y}$ and s^2 are called the **sufficient statistics** of the data, since they are sufficient to compute the MLE, without loss of information relative to using the raw data itself.

Note that you might be used to seeing the estimate for the variance written as

$$\hat{\sigma}_{\text{unb}}^2 = \frac{1}{N-1} \sum_{n=1}^N (y_n - \hat{\mu}_{\text{mle}})^2 \quad (4.38)$$

where we divide by $N-1$. This is not the MLE, but is a different kind of estimate, which happens to be unbiased (unlike the MLE); see Section 4.7.6.1 for details.⁴

4.2.6 Example: MLE for the multivariate Gaussian

In this section, we derive the maximum likelihood estimate for the parameters of a multivariate Gaussian.

First, let us write the log-likelihood, dropping irrelevant constants:

$$\ell(\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \log p(\mathcal{D} | \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{N}{2} \log |\boldsymbol{\Lambda}| - \frac{1}{2} \sum_{n=1}^N (\mathbf{y}_n - \boldsymbol{\mu})^\top \boldsymbol{\Lambda} (\mathbf{y}_n - \boldsymbol{\mu}) \quad (4.39)$$

where $\boldsymbol{\Lambda} = \boldsymbol{\Sigma}^{-1}$ is the **precision matrix** (inverse covariance matrix).

4. Note that, in Python, numpy defaults to the MLE, but Pandas defaults to the unbiased estimate, as explained in <https://stackoverflow.com/questions/24984178/different-std-in-pandas-vs-numpy/>.

4.2.6.1 MLE for the mean

Using the substitution $\mathbf{z}_n = \mathbf{y}_n - \boldsymbol{\mu}$, the derivative of a quadratic form (Equation (7.264)) and the chain rule of calculus, we have

$$\frac{\partial}{\partial \boldsymbol{\mu}} (\mathbf{y}_n - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{y}_n - \boldsymbol{\mu}) = \frac{\partial}{\partial \mathbf{z}_n} \mathbf{z}_n^\top \boldsymbol{\Sigma}^{-1} \mathbf{z}_n \frac{\partial \mathbf{z}_n}{\partial \boldsymbol{\mu}^\top} \quad (4.40)$$

$$= -1(\boldsymbol{\Sigma}^{-1} + \boldsymbol{\Sigma}^{-T}) \mathbf{z}_n \quad (4.41)$$

since $\frac{\partial \mathbf{z}_n}{\partial \boldsymbol{\mu}^\top} = -\mathbf{I}$. Hence

$$\frac{\partial}{\partial \boldsymbol{\mu}} \ell(\boldsymbol{\mu}, \boldsymbol{\Sigma}) = -\frac{1}{2} \sum_{n=1}^N -2\boldsymbol{\Sigma}^{-1} (\mathbf{y}_n - \boldsymbol{\mu}) = \boldsymbol{\Sigma}^{-1} \sum_{n=1}^N (\mathbf{y}_n - \boldsymbol{\mu}) = 0 \quad (4.42)$$

$$\hat{\boldsymbol{\mu}} = \frac{1}{N} \sum_{n=1}^N \mathbf{y}_n = \bar{\mathbf{y}} \quad (4.43)$$

So the MLE of $\boldsymbol{\mu}$ is just the empirical mean.

4.2.6.2 MLE for the covariance matrix

We can use the trace trick (Equation (7.36)) to rewrite the log-likelihood in terms of the precision matrix $\boldsymbol{\Lambda} = \boldsymbol{\Sigma}^{-1}$ as follows:

$$\ell(\hat{\boldsymbol{\mu}}, \boldsymbol{\Lambda}) = \frac{N}{2} \log |\boldsymbol{\Lambda}| - \frac{1}{2} \sum_n \text{tr}[(\mathbf{y}_n - \hat{\boldsymbol{\mu}})(\mathbf{y}_n - \hat{\boldsymbol{\mu}})^\top \boldsymbol{\Lambda}] \quad (4.44)$$

$$= \frac{N}{2} \log |\boldsymbol{\Lambda}| - \frac{1}{2} \text{tr}[\mathbf{S}_{\bar{\mathbf{y}}} \boldsymbol{\Lambda}] \quad (4.45)$$

$$\mathbf{S}_{\bar{\mathbf{y}}} \triangleq \sum_{n=1}^N (\mathbf{y}_n - \bar{\mathbf{y}})(\mathbf{y}_n - \bar{\mathbf{y}})^\top = \left(\sum_n \mathbf{y}_n \mathbf{y}_n^\top \right) - N \bar{\mathbf{y}} \bar{\mathbf{y}}^\top \quad (4.46)$$

where $\mathbf{S}_{\bar{\mathbf{y}}}$ is the **scatter matrix** centered on $\bar{\mathbf{y}}$.

We can rewrite the scatter matrix in a more compact form as follows:

$$\mathbf{S}_{\bar{\mathbf{y}}} = \tilde{\mathbf{Y}}^\top \tilde{\mathbf{Y}} = \mathbf{Y}^\top \mathbf{C}_N^\top \mathbf{C}_N \mathbf{Y} = \mathbf{Y}^\top \mathbf{C}_N \mathbf{Y} \quad (4.47)$$

where

$$\mathbf{C}_N \triangleq \mathbf{I}_N - \frac{1}{N} \mathbf{1}_N \mathbf{1}_N^\top \quad (4.48)$$

is the **centering matrix**, which converts $\mathbf{Y}$ to $\tilde{\mathbf{Y}}$ by subtracting the mean $\bar{\mathbf{y}} = \frac{1}{N} \mathbf{Y}^\top \mathbf{1}_N$ off every row.

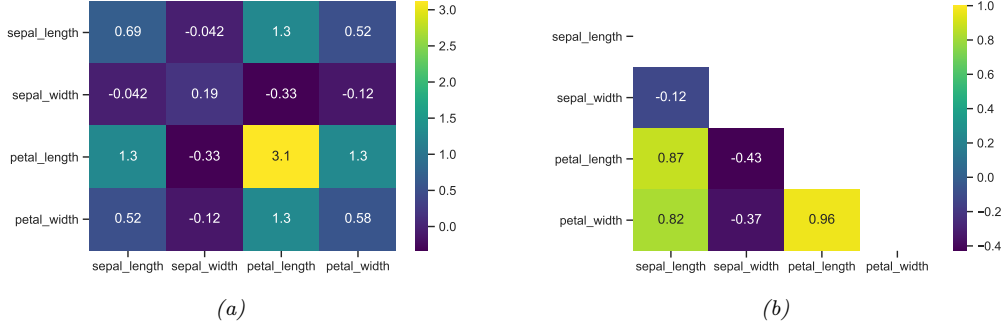


Figure 4.1: (a) Covariance matrix for the features in the iris dataset from Section 1.2.1.1. (b) Correlation matrix. We only show the lower triangle, since the matrix is symmetric and has a unit diagonal. Compare this to Figure 1.3. Generated by `iris_cov_mat.ipynb`.

Using results from Section 7.8, we can compute derivatives of the loss with respect to $\mathbf{\Lambda}$ to get

$$\frac{\partial \ell(\hat{\boldsymbol{\mu}}, \mathbf{\Lambda})}{\partial \mathbf{\Lambda}} = \frac{N}{2} \mathbf{\Lambda}^{-T} - \frac{1}{2} \mathbf{S}_{\bar{\mathbf{y}}}^T = \mathbf{0} \quad (4.49)$$

$$\mathbf{\Lambda}^{-T} = \mathbf{\Lambda}^{-1} = \mathbf{\Sigma} = \frac{1}{N} \mathbf{S}_{\bar{\mathbf{y}}} \quad (4.50)$$

$$\hat{\mathbf{\Sigma}} = \frac{1}{N} \sum_{n=1}^N (\mathbf{y}_n - \bar{\mathbf{y}})(\mathbf{y}_n - \bar{\mathbf{y}})^T = \frac{1}{N} \mathbf{Y}^T \mathbf{C}_N \mathbf{Y} \quad (4.51)$$

Thus the MLE for the covariance matrix is the empirical covariance matrix. See Figure 4.1a for an example.

Sometimes it is more convenient to work with the correlation matrix defined in Equation (3.8). This can be computed using

$$\text{corr}(\mathbf{Y}) = (\text{diag}(\mathbf{\Sigma}))^{-\frac{1}{2}} \mathbf{\Sigma} (\text{diag}(\mathbf{\Sigma}))^{-\frac{1}{2}} \quad (4.52)$$

where $\text{diag}(\mathbf{\Sigma})^{-\frac{1}{2}}$ is a diagonal matrix containing the entries $1/\sigma_i$. See Figure 4.1b for an example.

Note, however, that the MLE may overfit or be numerically unstable, especially when the number of samples N is small compared to the number of dimensions D . The main problem is that $\mathbf{\Sigma}$ has $O(D^2)$ parameters, so we may need a lot of data to reliably estimate it. In particular, as we see from Equation (4.51), the MLE for a full covariance matrix is singular if $N < D$. And even when $N > D$, the MLE can be ill-conditioned, meaning it is close to singular. We discuss solutions to this problem in Section 4.5.2.

4.2.7 Example: MLE for linear regression

We briefly mentioned linear regression in Section 2.6.3. Recall that it corresponds to the following model:

$$p(y|\mathbf{x}; \boldsymbol{\theta}) = \mathcal{N}(y|\mathbf{w}^T \mathbf{x}, \sigma^2) \quad (4.53)$$

significantly speed up the method; this is called **Hamiltonian Monte Carlo** or **HMC**. See the sequel to this book, [Mur23], for details.

4.7 Frequentist statistics *

The approach to statistical inference that we described in Section 4.6 is called Bayesian statistics. It treats parameters of models just like any other unknown random variable, and applies the rules of probability theory to infer them from data. Attempts have been made to devise approaches to statistical inference that avoid treating parameters like random variables, and which thus avoid the use of priors and Bayes rule. This alternative approach is known as **frequentist statistics**, **classical statistics** or **orthodox statistics**.

The basic idea (formalized in Section 4.7.1) is to represent uncertainty by calculating how a quantity estimated from data (such as a parameter or a predicted label) would change if the data were changed. It is this notion of variation across repeated trials that forms the basis for modeling uncertainty used by the frequentist approach. By contrast, the Bayesian approach views probability in terms of information rather than repeated trials. This allows the Bayesian to compute the probability of one-off events, as we discussed in Section 2.1.1. Perhaps more importantly, the Bayesian approach avoids certain paradoxes that plague the frequentist approach (see Section 4.7.5 and Section 5.5.4). These pathologies led the famous statistician George Box to say:

I believe that it would be very difficult to persuade an intelligent person that current [frequentist] statistical practice was sensible, but that there would be much less difficulty with an approach via likelihood and Bayes' theorem. — George Box, 1962 (quoted in [Jay76]).

Nevertheless, it is useful to be familiar with frequentist statistics, since it is widely used, and has some key concepts that are useful even for Bayesians [Rub84].

4.7.1 Sampling distributions

In frequentist statistics, uncertainty is not represented by the posterior distribution of a random variable, but instead by the **sampling distribution** of an **estimator**.

The term “estimator” is defined in the section on decision theory in Section 5.1, but in brief, an estimator $\delta : \mathcal{D} \rightarrow \mathcal{A}$ is a decision procedure that specifies what action to take given some observed data. The action could be to predict a class label, or the next observation, or to predict the unknown parameters. In the latter case, the estimator is often denoted by $\hat{\theta}$, but this notation is ambiguous, since it looks like it represents a parameter vector rather than a function. So instead we will use the notation $\hat{\Theta}$. This function could compute the MLE, or the method of moments estimate, etc. The output of this function, when applied to a specific dataset of size N , is denote $\hat{\theta} = \hat{\Theta}(\mathcal{D})$, where $\mathcal{D} = \{\mathbf{x}_1, \dots, \mathbf{x}_N\}$.

The key idea in frequentist statistics is to view the data $\mathcal{D}$ as a random variable, and the parameters from which the data are drawn, θ^* , as a fixed but unknown constant. Thus $\hat{\theta} = \hat{\Theta}(\mathcal{D})$ is a random variable, and its distribution is known as the sampling distribution of the estimator. To understand what this means, suppose we create S different datasets, each of the form

$$\mathcal{D}^{(s)} = \{\mathbf{x}_n \sim p(\mathbf{x}_n | \theta^*) : n = 1 : N\} \quad (4.217)$$

We denote this by $\mathcal{D}^{(s)} \sim \boldsymbol{\theta}^*$ for brevity. Now we apply the estimator to each $\mathcal{D}^{(s)}$ to get a set of estimates, $\{\hat{\boldsymbol{\theta}}(\mathcal{D}^{(s)})\}$. As we let $S \rightarrow \infty$, the distribution induced by this set is the sampling distribution of the estimator. More precisely, we have

$$\text{SamplingDist}(\hat{\boldsymbol{\Theta}}, \boldsymbol{\theta}^*) = \text{PushThrough}(p(\tilde{\mathcal{D}}|\boldsymbol{\theta}^*), \hat{\boldsymbol{\Theta}}) \quad (4.218)$$

where we push the data distribution through the estimator function to induce a distribution of estimates. In some cases, we can compute the sampling distribution analytically, as we discuss in Section 4.7.2, although typically we need to approximate it by Monte Carlo, as we discuss in Section 4.7.3.

4.7.2 Gaussian approximation of the sampling distribution of the MLE

The most common estimator is the MLE. When the sample size becomes large, the sampling distribution of the MLE for certain models becomes Gaussian. This is known as the **asymptotic normality** of the sampling distribution. More formally, we have the following result:

Theorem 4.7.1. *If the parameters are identifiable, then*

$$\text{SamplingDist}(\hat{\boldsymbol{\Theta}}^{\text{mle}}, \boldsymbol{\theta}^*) \rightarrow \mathcal{N}(\cdot|\boldsymbol{\theta}^*, (N\mathbf{F}(\boldsymbol{\theta}^*))^{-1}) \quad (4.219)$$

where $\mathbf{F}(\boldsymbol{\theta}^*)$ is the **Fisher information matrix**, defined in Equation (4.220).

Equivalently, the above result says that the distribution of $\sqrt{NF(\boldsymbol{\theta}^*)}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}^*)$ approaches $\mathcal{N}(\mathbf{0}, \mathbf{I})$, where $\hat{\boldsymbol{\theta}} = \hat{\boldsymbol{\Theta}}^{\text{mle}}(\tilde{\mathcal{D}})$.

The Fisher information matrix measures the amount of curvature of the log-likelihood surface at its peak, as we show below. More formally, the **Fisher information matrix (FIM)** is defined to be the covariance of the gradient of the log likelihood (also called the **score function**):

$$\mathbf{F}(\boldsymbol{\theta}) \triangleq \mathbb{E}_{\mathbf{x} \sim p(\mathbf{x}|\boldsymbol{\theta})} [\nabla \log p(\mathbf{x}|\boldsymbol{\theta}) \nabla \log p(\mathbf{x}|\boldsymbol{\theta})^\top] \quad (4.220)$$

Hence the (i, j) 'th entry has the form

$$F_{ij} = \mathbb{E}_{\mathbf{x} \sim \boldsymbol{\theta}} \left[\left(\frac{\partial}{\partial \theta_i} \log p(\mathbf{x}|\boldsymbol{\theta}) \right) \left(\frac{\partial}{\partial \theta_j} \log p(\mathbf{x}|\boldsymbol{\theta}) \right) \right] \quad (4.221)$$

One can show the following result.

Theorem 4.7.2. *If $\log p(\mathbf{x}|\boldsymbol{\theta})$ is twice differentiable, and under certain regularity conditions, the FIM is equal to the expected Hessian of the NLL, i.e.,*

$$\mathbf{F}_{ij} = -\mathbb{E}_{\mathbf{x} \sim \boldsymbol{\theta}} \left[\frac{\partial^2}{\partial \theta_i \partial \theta_j} \log p(\mathbf{x}|\boldsymbol{\theta}) \right] \quad (4.222)$$

If we replace the expectation over $\mathbf{x}$ with the observed value, to get the empirical FIM, we see this is equal to the Hessian of the NLL. This helps us understand the result in Equation (4.219): a log-likelihood function with high curvature (large Hessian) will result in a low variance estimate, since the parameters are “well determined” by the data, and hence robust to repeated sampling.

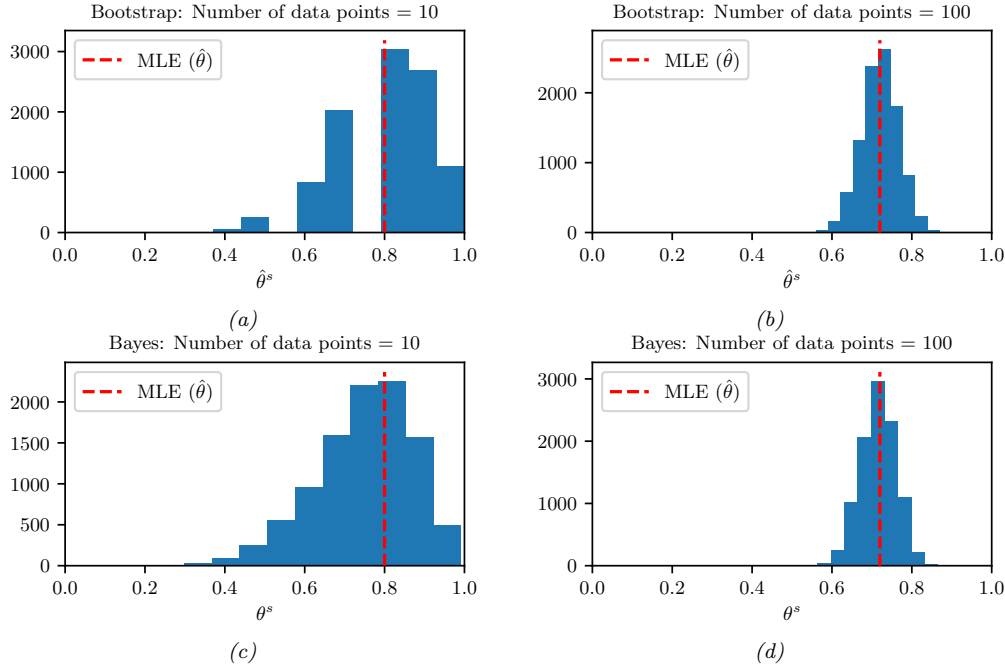


Figure 4.23: Bootstrap (top row) vs Bayes (bottom row). The N data cases were generated from $\text{Ber}(\theta = 0.7)$. Left column: $N = 10$. Right column: $N = 100$. (a-b) A bootstrap approximation to the sampling distribution of the MLE for a Bernoulli distribution. We show the histogram derived from $B = 10,000$ bootstrap samples. (c-d) Histogram of 10,000 samples from the posterior distribution using a uniform prior. Generated by [bootstrap_demo_bernoulli.ipynb](#).

In the scalar case, we have that $\mathbb{V}[\hat{\theta} - \theta^*] \rightarrow \frac{1}{NF(\theta^*)}$. The square root of the variance of the sampling distribution of an estimator is known as its **standard error** or se. Hence we can say that the distribution of $\frac{\hat{\theta} - \theta^*}{\text{se}}$ approaches $\mathcal{N}(0, 1)$. In practice the se is not known, but it can be estimated from data. For example, suppose $X_n \sim \text{Ber}(\theta^*)$ and let $\hat{\theta} = \frac{1}{N} \sum_{n=1}^N X_n$ be the MLE. The standard error is $\text{se} = \sqrt{\mathbb{V}[\hat{\theta}]} = \sqrt{\theta^*(1 - \theta^*)/N}$, so the estimated standard error is $\hat{\text{se}} = \sqrt{\hat{\theta}(1 - \hat{\theta})/N}$.

4.7.3 Bootstrap approximation of the sampling distribution of any estimator

In cases where the estimator is a complex function of the data (e.g., not just an MLE), or when the sample size is small, we can approximate its sampling distribution using a Monte Carlo technique known as the **bootstrap**.

The idea is simple. If we knew the true parameters θ^* , we could generate many (say S) fake datasets, each of size N , from the true distribution, using $\tilde{\mathcal{D}}^{(s)} = \{\mathbf{x}_n \sim p(\mathbf{x}_n | \theta^*) : n = 1 : N\}$. We could then compute our estimate from each sample, $\hat{\theta}^s = \hat{\theta}(\tilde{\mathcal{D}}^{(s)})$ and use the empirical distribution of the resulting $\hat{\theta}^s$ as our estimate of the sampling distribution, as in Equation (4.218). Since θ^* is

unknown, we can use the dataset itself as an empirical approximation to $p(\mathbf{x}_n|\boldsymbol{\theta}^*)$. More precisely, the idea is to generate each $\tilde{D}^{(s)}$ by sampling N data points with replacement from the original dataset. (If we sampled N times without replacement, we would just recover the original dataset.) This is like “lifting yourself up from your own bootstraps”, since we use the observed data sample to make new hypothetical data samples.⁶

Figure 4.23(a-b) shows an example where we compute the sampling distribution of the MLE for a Bernoulli using the bootstrap. When $N = 10$, we see that the sampling distribution is asymmetric, and therefore quite far from Gaussian, but when $N = 100$, the distribution looks more Gaussian, as theory suggests (see Section 4.7.2).

The number of unique data points in a bootstrap sample is just $0.632 \times N$, on average. (To see this, note that the probability an item is picked at least once is $(1 - (1 - 1/N)^N)$, which approaches $1 - e^{-1} \approx 0.632$ for large N .) However, there are more sophisticated versions of bootstrap that improve on this (see e.g., [Efr87; EH16]).

4.7.3.1 Bootstrap is a “poor man’s” posterior

A natural question is: what is the connection between the parameter estimates $\hat{\boldsymbol{\theta}}^s = \hat{\boldsymbol{\theta}}(\mathcal{D}^{(s)})$ computed by the bootstrap and parameter values sampled from the posterior, $\boldsymbol{\theta}^s \sim p(\cdot|\mathcal{D})$? Conceptually they are quite different. But in the common case that the estimator is MLE and the prior is not very strong, they can be quite similar. For example, Figure 4.23(c-d) shows an example where we compute the posterior using a uniform Beta(1,1) prior, and then sample from it. We see that the posterior and the sampling distribution are quite similar. So one can think of the bootstrap distribution as a “poor man’s” posterior [HTF01, p235].

However, perhaps surprisingly, bootstrap can be slower than posterior sampling. The reason is that the bootstrap has to generate S sampled datasets, and then fit a model to each one. By contrast, in posterior sampling, we only have to “fit” a model once given a single dataset. (Some methods for speeding up the bootstrap when applied to massive data sets are discussed in [Kle+11].)

4.7.4 Confidence intervals

In frequentist statistics, we use the variability induced by the sampling distribution as a way to estimate uncertainty of a parameter estimate.

In particular, we define a $100(1 - \alpha)\%$ **confidence interval** for parameter θ as an estimator that returns an interval that captures the true parameter with probability at least $1 - \alpha$. Denote the estimator by $I(\mathcal{D}) = (\ell(\mathcal{D}), u(\mathcal{D}))$. The sampling distribution of this estimator is the distribution that is induced by sampling $\tilde{\mathcal{D}} \sim \boldsymbol{\theta}^*$ and then computing $I(\tilde{\mathcal{D}})$. We require that

$$\Pr(\theta^* \in I(\tilde{\mathcal{D}}) | \tilde{\mathcal{D}} \sim \boldsymbol{\theta}^*) \geq 1 - \alpha \quad (4.223)$$

It is common to set $\alpha = 0.05$, which yields a 95% CI. This means that, if we repeatedly sampled data, and compute $I(\tilde{\mathcal{D}})$ for each such dataset, then about 95% of such intervals will contain the true parameter θ .

6. This is called the **non-parametric bootstrap**. There is another variant, called the **parametric bootstrap**, in which we sample each $\tilde{D}^{(s)}$ from $p(\mathbf{x}_n|\hat{\boldsymbol{\theta}}(\mathcal{D}))$; this requires fitting a parametric model to the original data and then sampling from it.

Let us give an example. Suppose that $\hat{\theta} = \hat{\Theta}(\mathcal{D})$ is an estimator for some parameter with true but unknown value θ^* . Also, suppose that the sampling distribution of $\Delta = \theta^* - \hat{\theta}$ is known. Let $\underline{\delta}$ and $\bar{\delta}$ denote its $\alpha/2$ and $1 - \alpha/2$ quantiles, so

$$\Pr(\underline{\delta} \leq \Delta \leq \bar{\delta}) = \Pr(\underline{\delta} \leq \theta^* - \hat{\theta} \leq \bar{\delta}) = 1 - \alpha \quad (4.224)$$

Rearranging we get

$$\Pr(\hat{\theta} + \underline{\delta} \leq \theta^* \leq \hat{\theta} + \bar{\delta}) = 1 - \alpha \quad (4.225)$$

Hence we can construct a $100(1 - \alpha)\%$ confidence interval as follows:

$$I(\mathcal{D}) = (L, U) = (\hat{\theta}(\mathcal{D}) + \underline{\delta}(\mathcal{D}), \hat{\theta}(\mathcal{D}) + \bar{\delta}(\mathcal{D})) \quad (4.226)$$

In some cases, we can analytically compute the sampling distribution of the above interval. However, it is more common to assume a Gaussian approximation to the sampling distribution, as in Section 4.7.2. In this case, we have $\hat{\theta} \approx \mathcal{N}(\theta^*, \hat{\text{se}}^2)$. Hence we can compute an approximate CI using

$$I = (\hat{\theta} - z_{\alpha/2} \hat{\text{se}}, \hat{\theta} + z_{\alpha/2} \hat{\text{se}}) \quad (4.227)$$

where $z_{\alpha/2}$ is the $\alpha/2$ quantile of the Gaussian cdf. If we set $\alpha = 0.05$, we have $z_{\alpha/2} = 1.96$, which justifies the common approximation $\hat{\theta} \pm 2\hat{\text{se}}$.

If the Gaussian approximation is not a good one, we can use a bootstrap approximation (see Section 4.7.3). In particular, we sample S datasets from $\hat{\theta}(\mathcal{D})$, and apply the estimator to each one to get $\hat{\theta}(\mathcal{D}^{(s)})$; we then use the empirical distribution of $\hat{\theta}(\mathcal{D}) - \hat{\theta}(\mathcal{D}^{(s)})$ as an approximation to the sampling distribution of Δ . We can then use the $\alpha/2$ and $1 - \alpha/2$ quantiles of this distribution to derive the CI (see [Was04, p110] for details).

4.7.5 Caution: Confidence intervals are not credible

It is commonly believed that a 95% confidence interval I for a parameter estimate θ given data $\mathcal{D}$ means that the true parameter lies in this interval with probability 0.95, i.e., $p(\theta^* \in I|\mathcal{D}) = 0.95$. However, this quantity is what a Bayesian credible interval computes (Section 4.6.6), but is not what a frequentist confidence interval computes. Instead the frequentist approach just means that the procedure for generating CIs will contain the true value 95% of the time. That is, if we repeatedly sample datasets $\tilde{\mathcal{D}}$ from θ^* , and compute their CIs to get $I(\tilde{\mathcal{D}})$, then we have $\Pr(\theta^* \in I(\tilde{\mathcal{D}})) = 0.95$, as we explain in Section 4.7.4. Thus we see that these concepts are quite different: In the frequentist approach, θ is treated as an unknown fixed constant, and the data is treated as random. In the Bayesian approach, we treat the data as fixed (since it is known) and the parameter as random (since it is unknown).

This counter-intuitive definition of confidence intervals can lead to bizarre results. Consider the following example from [Ber85, p11]. Suppose we draw two integers $\mathcal{D} = (y_1, y_2)$ from

$$p(y|\theta) = \begin{cases} 0.5 & \text{if } y = \theta \\ 0.5 & \text{if } y = \theta + 1 \\ 0 & \text{otherwise} \end{cases} \quad (4.228)$$

If $\theta = 39$, we would expect the following outcomes each with probability 0.25:

$$(39, 39), (39, 40), (40, 39), (40, 40) \quad (4.229)$$

Let $m = \min(y_1, y_2)$ and define the following interval:

$$[\ell(\mathcal{D}), u(\mathcal{D})] = [m, m] \quad (4.230)$$

For the above samples this yields

$$[39, 39], [39, 39], [39, 39], [40, 40] \quad (4.231)$$

Hence Equation (4.230) is clearly a 75% CI, since 39 is contained in 3/4 of these intervals. However, if we observe $\mathcal{D} = (39, 40)$ then $p(\theta = 39|\mathcal{D}) = 1.0$, so we know that θ must be 39, yet we only have 75% “confidence” in this fact. We see that the CI will “cover” the true parameter 75% of the time, if we compute multiple CIs from different randomly sampled datasets, but if we just have a single observed dataset, and hence a single CI, then the frequentist “coverage” probability can be very misleading.

Another, less contrived, example is as follows. Suppose we want to estimate the parameter θ of a Bernoulli distribution. Let $\bar{y} = \frac{1}{N} \sum_{n=1}^N y_n$ be the sample mean. The MLE is $\hat{\theta} = \bar{y}$. An approximate 95% confidence interval for a Bernoulli parameter is $\bar{y} \pm 1.96\sqrt{\bar{y}(1-\bar{y})/N}$ (this is called a **Wald interval** and is based on a Gaussian approximation to the Binomial distribution; compare to Equation (4.128)). Now consider a single trial, where $N = 1$ and $y_1 = 0$. The MLE is 0, which overfits, as we saw in Section 4.5.1. But our 95% confidence interval is also $(0, 0)$, which seems even worse. It can be argued that the above flaw is because we approximated the true sampling distribution with a Gaussian, or because the sample size was too small, or the parameter “too extreme”. However, the Wald interval can behave badly even for large N , and non-extreme parameters [BCD01]. By contrast, a Bayesian credible interval with a non-informative Jeffreys prior behaves in the way we would expect.

Several more interesting examples, along with Python code, can be found at [Van14]. See also [Hoe+14; Mor+16; Lyu+20; Cha+19b], who show that many people, including professional statisticians, misunderstand and misuse frequentist confidence intervals in practice, whereas Bayesian credible intervals do not suffer from these problems.

4.7.6 The bias-variance tradeoff

An estimator is a procedure applied to data which returns an estimand. Let $\hat{\theta}()$ be the estimator, and $\hat{\theta}(\mathcal{D})$ be the estimand. In frequentist statistics, we treat the data as a random variable, drawn from some true but unknown distribution, $p^*(\mathcal{D})$; this induces a distribution over the estimand, $p^*(\hat{\theta}(\mathcal{D}))$, known as the sampling distribution (see Section 4.7.1). In this section, we discuss two key properties of this distribution, its bias and its variance, which we define below.

4.7.6.1 Bias of an estimator

The **bias** of an estimator is defined as

$$\text{bias}(\hat{\theta}(\cdot)) \triangleq \mathbb{E} [\hat{\theta}(\mathcal{D})] - \theta^* \quad (4.232)$$

where θ^* is the true parameter value, and the expectation is wrt “nature’s distribution” $p(\mathcal{D}|\theta^*)$. If the bias is zero, the estimator is called **unbiased**. For example, the MLE for a Gaussian mean is unbiased:

$$\text{bias}(\hat{\mu}) = \mathbb{E}[\bar{x}] - \mu = \mathbb{E}\left[\frac{1}{N} \sum_{n=1}^N x_n\right] - \mu = \frac{N\mu}{N} - \mu = 0 \quad (4.233)$$

where $\bar{x}$ is the sample mean.

However, the MLE for a Gaussian variance, $\sigma_{\text{mle}}^2 = \frac{1}{N} \sum_{n=1}^N (x_n - \bar{x})^2$, is not an unbiased estimator of σ^2 . In fact, one can show (Exercise 4.7) that

$$\mathbb{E}[\sigma_{\text{mle}}^2] = \frac{N-1}{N} \sigma^2 \quad (4.234)$$

so the ML estimator slightly underestimates the variance. Intuitively, this is because we “use up” one of the data points to estimate the mean, so if we have a sample size of 1, we will estimate the variance to be 0. If, however, μ is known, the ML estimator is unbiased (see Exercise 4.8).

Now consider the following estimator

$$\sigma_{\text{unb}}^2 \triangleq \frac{1}{N-1} \sum_{n=1}^N (x_n - \bar{x})^2 = \frac{N}{N-1} \sigma_{\text{mle}}^2 \quad (4.235)$$

This is an unbiased estimator, which we can easily prove as follows:

$$\mathbb{E}[\sigma_{\text{unb}}^2] = \frac{N}{N-1} \mathbb{E}[\sigma_{\text{mle}}^2] = \frac{N}{N-1} \frac{N-1}{N} \sigma^2 = \sigma^2 \quad (4.236)$$

4.7.6.2 Variance of an estimator

It seems intuitively reasonable that we want our estimator to be unbiased. However, being unbiased is not enough. For example, suppose we want to estimate the mean of a Gaussian from $\mathcal{D} = \{x_1, \dots, x_N\}$. The estimator that just looks at the first data point, $\hat{\theta}(\mathcal{D}) = x_1$, is an unbiased estimator, but will generally be further from θ^* than the empirical mean $\bar{x}$ (which is also unbiased). So the variance of an estimator is also important.

We define the variance of an estimator as follows:

$$\mathbb{V}[\hat{\theta}] \triangleq \mathbb{E}[\hat{\theta}^2] - \left(\mathbb{E}[\hat{\theta}]\right)^2 \quad (4.237)$$

where the expectation is taken wrt $p(\mathcal{D}|\theta^*)$. This measures how much our estimate will change as the data changes. We can extend this to a covariance matrix for vector valued estimators.

Intuitively we would like the variance of our estimator to be as small as possible. Therefore, a natural question is: how low can the variance go? A famous result, called the **Cramer-Rao lower bound**, provides a lower bound on the variance of any unbiased estimator. More precisely, let $X_1, \dots, X_N \sim p(X|\theta^*)$ and $\hat{\theta} = \hat{\theta}(x_1, \dots, x_N)$ be an unbiased estimator of θ^* . Then, under various smoothness assumptions on $p(X|\theta^*)$, we have $\mathbb{V}[\hat{\theta}] \geq \frac{1}{NF(\theta^*)}$, where $F(\theta^*)$ is the Fisher information matrix (Section 4.7.2). A proof can be found e.g., in [Ric95, p275].

It can be shown that the MLE achieves the Cramer Rao lower bound, and hence has the smallest asymptotic variance of any unbiased estimator. Thus MLE is said to be **asymptotically optimal**.

4.7.6.3 The bias-variance tradeoff

In this section, we discuss a fundamental tradeoff that needs to be made when picking a method for parameter estimation, assuming our goal is to minimize the mean squared error (MSE) of our estimate. Let $\hat{\theta} = \hat{\theta}(\mathcal{D})$ denote the estimate, and $\bar{\theta} = \mathbb{E}[\hat{\theta}]$ denote the expected value of the estimate (as we vary $\mathcal{D}$). (All expectations and variances are wrt $p(\mathcal{D}|\theta^*)$, but we drop the explicit conditioning for notational brevity.) Then we have

$$\mathbb{E}[(\hat{\theta} - \theta^*)^2] = \mathbb{E}\left[\left((\hat{\theta} - \bar{\theta}) + (\bar{\theta} - \theta^*)\right)^2\right] \quad (4.238)$$

$$= \mathbb{E}\left[(\hat{\theta} - \bar{\theta})^2\right] + 2(\bar{\theta} - \theta^*)\mathbb{E}[\hat{\theta} - \bar{\theta}] + (\bar{\theta} - \theta^*)^2 \quad (4.239)$$

$$= \mathbb{E}\left[(\hat{\theta} - \bar{\theta})^2\right] + (\bar{\theta} - \theta^*)^2 \quad (4.240)$$

$$= \mathbb{V}[\hat{\theta}] + \text{bias}^2(\hat{\theta}) \quad (4.241)$$

In words,

$$\text{MSE} = \text{variance} + \text{bias}^2$$

(4.242)

This is called the **bias-variance tradeoff** (see e.g., [GBD92]). What it means is that it might be wise to use a biased estimator, so long as it reduces our variance by more than the square of the bias, assuming our goal is to minimize squared error.

4.7.6.4 Example: MAP estimator for a Gaussian mean

Let us give an example, based on [Hof09, p79]. Suppose we want to estimate the mean of a Gaussian from $\mathbf{x} = (x_1, \dots, x_N)$. We assume the data is sampled from $x_n \sim \mathcal{N}(\theta^* = 1, \sigma^2)$. An obvious estimate is the MLE. This has a bias of 0 and a variance of

$$\mathbb{V}[\bar{x}|\theta^*] = \frac{\sigma^2}{N} \quad (4.243)$$

But we could also use a MAP estimate. In Section 4.6.4.2, we show that the MAP estimate under a Gaussian prior of the form $\mathcal{N}(\theta_0, \sigma^2/\kappa_0)$ is given by

$$\tilde{x} \triangleq \frac{N}{N + \kappa_0}\bar{x} + \frac{\kappa_0}{N + \kappa_0}\theta_0 = w\bar{x} + (1 - w)\theta_0 \quad (4.244)$$

where $0 \leq w \leq 1$ controls how much we trust the MLE compared to our prior. The bias and variance are given by

$$\mathbb{E}[\tilde{x}] - \theta^* = w\theta^* + (1 - w)\theta_0 - \theta^* = (1 - w)(\theta_0 - \theta^*) \quad (4.245)$$

$$\mathbb{V}[\tilde{x}] = w^2 \frac{\sigma^2}{N} \quad (4.246)$$

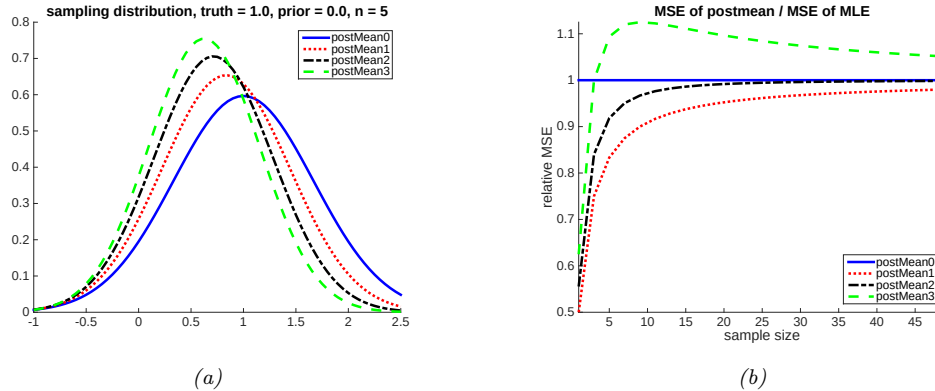


Figure 4.24: Left: Sampling distribution of the MAP estimate (equivalent to the posterior mean) under a $\mathcal{N}(\theta_0 = 0, \sigma^2/\kappa_0)$ prior with different prior strengths κ_0 . (If we set $\kappa = 0$, the MAP estimate reduces to the MLE.) The data is $n = 5$ samples drawn from $\mathcal{N}(\theta^* = 1, \sigma^2 = 1)$. Right: MSE relative to that of the MLE versus sample size. Adapted from Figure 5.6 of [Hof09]. Generated by [samplingDistributionGaussianShrinkage.ipynb](#).

So although the MAP estimate is biased (assuming $w < 1$), it has lower variance.

Let us assume that our prior is slightly misspecified, so we use $\theta_0 = 0$, whereas the truth is $\theta^* = 1$. In Figure 4.24(a), we see that the sampling distribution of the MAP estimate for $\kappa_0 > 0$ is biased away from the truth, but has lower variance (is narrower) than that of the MLE.

In Figure 4.24(b), we plot $\text{mse}(\tilde{x})/\text{mse}(\bar{x})$ vs N . We see that the MAP estimate has lower MSE than the MLE for $\kappa_0 \in \{1, 2\}$. The case $\kappa_0 = 0$ corresponds to the MLE, and the case $\kappa_0 = 3$ corresponds to a strong prior, which hurts performance because the prior mean is wrong. Thus we see that, provided the prior strength is properly “tuned”, a MAP estimate can outperform an ML estimate in terms of minimizing MSE.

4.7.6.5 Example: MAP estimator for linear regression

Another important example of the bias-variance tradeoff arises in ridge regression, which we discuss in Section 11.3. In brief, this corresponds to MAP estimation for linear regression under a Gaussian prior, $p(\mathbf{w}) = \mathcal{N}(\mathbf{w}|\mathbf{0}, \lambda^{-1}\mathbf{I})$. The zero-mean prior encourages the weights to be small, which reduces overfitting; the precision term, λ , controls the strength of this prior. Setting $\lambda = 0$ results in the MLE; using $\lambda > 0$ results in a biased estimate. To illustrate the effect on the variance, consider a simple example where we fit a 1d ridge regression model using 2 different values of λ . Figure 4.25 on the left plots each individual fitted curve, and on the right plots the average fitted curve. We see that as we increase the strength of the regularizer, the variance decreases, but the bias increases.

See also Figure 4.26 where we give a cartoon sketch of the bias variance tradeoff in terms of model complexity.

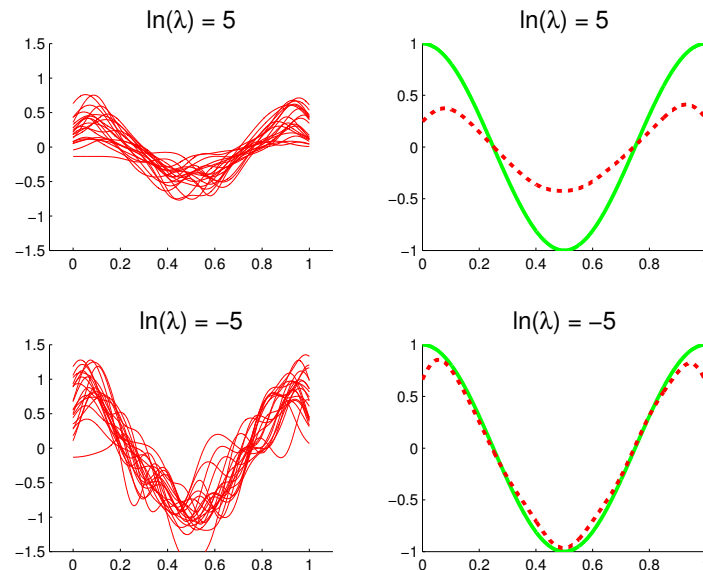


Figure 4.25: Illustration of bias-variance tradeoff for ridge regression. We generate 100 data sets from the true function, shown in solid green. Left: we plot the regularized fit for 20 different data sets. We use linear regression with a Gaussian RBF expansion, with 25 centers evenly spread over the $[0, 1]$ interval. Right: we plot the average of the fits, averaged over all 100 datasets. Top row: strongly regularized: we see that the individual fits are similar to each other (low variance), but the average is far from the truth (high bias). Bottom row: lightly regularized: we see that the individual fits are quite different from each other (high variance), but the average is close to the truth (low bias). Adapted from [Bis06] Figure 3.5. Generated by `biasVarModelComplexity3.ipynb`.

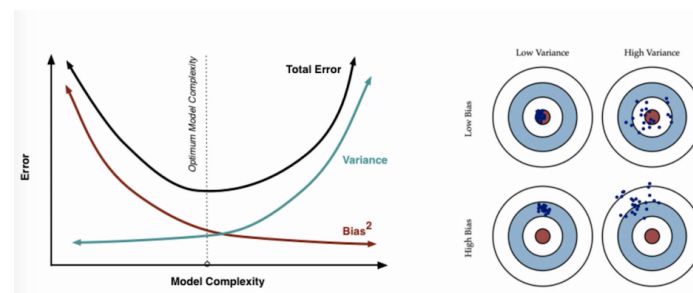


Figure 4.26: Cartoon illustration of the bias variance tradeoff. From <http://scott.fortmann-roe.com/docs/BiasVariance.html>. Used with kind permission of Scott Fortmann-Roe.